A PERFECT PROPER PUSHFORWARD OF A PERFECT COMPLEX ON A NOETHERIAN SPACE IS PERFECT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 11093-11099 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 11101-11114, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side; schemes twin uncounted per ruling.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Derived Categories of Spaces, lemmas perfect-direct-image and tor-dimension-rel; More on Morphisms, lemma check-perfect-stalks.

Source reference. Original lemma 069C: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11114–11114. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08IS: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11114–11114. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DK7: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11114–11114. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let Y be a Noetherian algebraic space over S. Let $f : X \to Y$ be a perfect proper morphism of algebraic spaces. Let $E \in D(\mathcal{O}_X)$ be perfect. Then Rf_*E is a perfect object of $D(\mathcal{O}_Y)$.*

Proof. We claim that Derived Categories of Spaces, Lemma 08IS applies. Conditions (1) and (2) are immediate. Condition (3) is local on X. Thus we may assume X and Y affine and E represented by a strictly perfect complex of $\mathcal{O}_X$-modules. Thus it suffices to show that $\mathcal{O}_X$ has finite tor dimension as a sheaf of $f^{-1}\mathcal{O}_Y$-modules on the étale site. By Derived Categories of Spaces, Lemma 0DK7 it suffices to check this on the Zariski site. This is equivalent to being perfect for finite type morphisms of schemes by More on Morphisms, Lemma 069C. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.